CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (IT/CE)

May-2014

IT307/IT307.01 Software Engineering (S.E.)

Date: 13.05.2014, Tuesday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

1. What are the unique challenges and elements that are involved in managing IT projects? How is software engineering useful in controlling these challenges? 4
2. What do you mean by milestones and deliverable? What are the contents that are intended into the deliverable? 3
3. Show how the failure curve of Software differs from that of Hardware. Software doesn't wear out but it deteriorates due to change. Justify. How do software myths affect a software project? 4

Q-2

[A] Name the risk based software development process model. Compare its advantages and disadvantages in keeping eyes with all other models. 4

[B] Quality, Reliability and Safety are related concepts, but are fundamentally different in a number ways. Discuss. Also explain why software reliability always takes precedence over efficiency. 4

OR

[B] Suggest and explain suitable process model that can be used to develop large software project. What are the major strengths of object oriented approach in developing large software project? 4

[C] Why is it difficult to gain a clear understanding of what the customer wants? Explain different steps of requirement engineering process. Describe any two requirement elicitation techniques. What can be done, if the requirements are changing continuously? 4

OR

[C] What is a risk? What are risk management activities? What types of risks are likely to encounter as software is built? How would you identify the risks and how to containment risks from a project! 4

Q-3

[A] Develop an E-R diagram and prepare Data Dictionary for Library management system. 4

[B] List out the different characteristics of good UI Design. List the desirable characteristics that a good user interface should possess. 4

OR

[B] When are verification and validation performed during the software life cycle? What do you mean by TQM and explain any four key elements of TQM? 4

[C] What is the role UML in framing architecture of the software? Briefly explain the general activities in performing Object Oriented Analysis (OOA) in UML. What are the 3 additional design and implantation models offered by UML? Define each of them. 4

SECTION-II

Q-4

1. What do you understand by black box testing and white box testing? Explain Equivalence Class Partitioning and Boundary value analysis and compare them. Consider a program which computes the square root of an input integer between 0 and 5000. Determine the equivalence class test cases. Determine the test cases using boundary value analysis also. 4
2. What is a component? Explain object oriented view of a component in brief. Explain the concept of Component Based Software Engineering (CBSD). Also explain the role of Agile (process model) in it. How is Agile differ from other process model? 4
3. What do the by coding guideline? Represents at least 4 coding guidelines. 3

Q-5

[A] Explain the need for software measures. Explain how LOC metric is reacted to FP based metric. Justify by giving example that FPs give better estimate of Software project size than LOC. 4

[B] List important attributes of Quality. How is software quality evaluated? What is SQA (Software Quality Assurance)? List and explain various SQA activities. 4

[C] Justly: Learning effective project management techniques will allow you to complete projects on time and bring them in under budget. What is project management plan? Explain the four major sections that form the software project management plan (SPMP). 4

OR

Q-5

[A] What do you mean by software documentation? Why is it important to properly document a software product? What are the different types of documents that need to be developed or produced? 4

[B] What do the by Configuration Management? How does it help to ensure high quality of a software product? Also explain: What are the different types of maintenance that a software product might need? 4

[C] Define Software Re-Engineering? What are the main objectives of Re-Engineering? How is Software Re-engineering differ from Software Reverse Engineering? Discuss a process model suitable and used for software re-engineering. 4

Q-6

[A] What do you mean by CASE? List out the benefits of CASE and CASE tools. 2

[B] What is the use of SEI CMM in software oranganizations? Explain all levels achieved in SEI CMM with their KPAs. 4

OR

[B] What do the by Configuration Management? How does it help to ensure high quality of a software product? What are the different types of maintenance that a software product might need? 4

[C] What is FTR? If your company is going to conduct a review meeting, who should be on the review committee and why? Describe design walk through and critical design review. Define and explain any three aspects of design and code reviews. Give an example of the types of errors detected during code inspection, design walk through, critical design review and code walkthrough. 6

OR

[C] What are the objectives and aim(s) of software testing? What are the attributes which will impact the testing process? What are the techniques of testing? What is the sequence of testing? What is cycloramatic complexity? How to calculate it using graph matrices. Show it with example. 6